

I. SECTION CAPACITY / LINE CAPACITY

1. Definition

→ “Number of trains that can be run each way on a given section of the railway in 24 hours.”

- **Single Line:** Total trains per direction in 24 hours
- **Double Line:** Capacity assessed **separately for Up and Down** directions
- This governs **train planning, regulation, timetabling, throughput, and policy decisions.**

2. Types of Line Capacity

Section capacity is expressed in **three technical measures:**

A. Maximum Line Capacity (Theoretical Capacity)

 *Definition:*

Maximum number of train paths that can be plotted in the **Master Chart** such that **no additional train** can be inserted.

B. Practical Line Capacity (Effective Line Capacity)

 *Definition:*

Number of trains that can actually be **scheduled and run safely and reliably** after providing **time allowances** for:

- Track / OHE / Signalling maintenance
- Operational irregularities
- Pathing margin
- Crossing / precedence requirement (especially on Single Line)
- Engineering blocks
- Unforeseen delays
- Crew/loco/yard constraints

- Running time of the **slowest** train on that section
- Used because the slowest train determines the **minimum headway**
- Includes **acceleration + deceleration + permanent speed restrictions**

C. Block Operation Time (t)

Includes:

- Time to obtain Line Clear
- Time for releasing block section
- Station operational delays
- S&T response time

Generally: **4–6 minutes** (varies with type of block instrument)

D. Efficiency Factor (e)

Efficiency factor (0.80 to 0.90) includes:

- Engineering blocks
- Irregularities
- Maintenance windows
- Signal failures
- Crew & loco constraints
- Pathing margins

IR commonly uses **0.85** as a practical value.

4. Sample (Exam-Style) Calculation

Given:

- Slowest train running time (**T**) = 30 minutes
- Block operation time (**t**) = 6 minutes
- Efficiency factor (**e**) = 0.85

$$C = \frac{1440 \times 0.85}{30 + 6} C = \frac{1224}{36} = 34 \text{ trains per day (approx.)}$$

Capacity per direction = 34 trains/day

(for single line $\rightarrow$ each way; for double line $\rightarrow$ per direction)

7. One-Line Summary for Memory

Scott's Formula gives the number of trains/day using the slowest train's running time + block clearance time, multiplied by efficiency, over 1440 minutes.

CALCULATION OF LINE CAPACITY — CHARTING METHOD

1. What is the Charting Method?

The **Charting Method** uses a **time–distance graph (Master Chart)** plotted for **24 hours**, representing the actual movement of trains over a section.

It is considered the **most reliable and practical** method for determining **true, usable section capacity**.

- ✓ Used by almost all Zonal Railways
- ✓ Reflects real operational conditions
- ✓ More accurate than mathematical formulas (Scott, CPRO, etc.)

2. Steps in Charting Method (Exam-Oriented)

Step 1: Draw a Master Chart

- X-axis → **Time (24 hours or 1440 minutes)**
- Y-axis → **Distance between stations** (block stations, block sections)
- Represents the **complete section** under study.

Step 2: Plot All Scheduled Coaching Train Paths

- Plot **all coaching trains first**, as their timings are fixed by timetable
 - Include:
 - Mail/Express
 - Passenger
 - DEMU/MEMU
 - Intercity / Suburban services
 - Each train path is drawn as a **sloping line** representing movement through various block stations.
- ✓ This forms the **base skeleton** of the master chart.

Step 3: Insert Goods Train Paths

- After coaching trains, **insert maximum possible goods train paths**
 - Up direction
 - Down direction
- Consider:
 - Running time
 - Acceleration/deceleration
 - Crossings and precedence (especially on Single Line)
 - Terminal detentions
 - Loop line lengths
 - Permissible speeds
 - Time for locomotive/crew change
 - OHE/Signal constraints

✓ Continue inserting until **no additional train path can be accommodated** without violating rules of operation.

Step 4: Achieving “Saturation Point”

The process continues until:

- No more train path can be plotted **without causing conflict**
- No additional train can pass safely through block sections
- All operational limitations (block length, station spacing, loop availability) are respected

✂ **This saturation point = Maximum Section Capacity**

3. What Does Charting Method Consider?

Unlike formula-based methods, the Charting Method **captures real-world operational factors**, such as:

A. Block Section Limitations

- Longest block section → critical
- Location of crossing stations
- Interlocking arrangements

- Platform availability
- Yard capacity
- Loco change facilities
- Crew changing points
- Coaching and goods terminals

- Slow vs fast trains
- Goods vs coaching movements
- Priority trains requiring non-stop runs

- Precedence
- Detention
- Engineering blocks
- Mandatory maintenance allowances
- OHE/Signal failure margins
- LC gate operations

✓ This makes it **closest to actual working capacity.**

MEASURES TO INCREASE SECTION CAPACITY WITHOUT INCURRING EXPENDITURE

(Purely operational & managerial improvements — NO capital investment required)

Increasing section capacity **does not always require physical augmentation** like doubling, electrification, or signaling upgrades.

A large portion of capacity improvement can be achieved through **better planning, control, and management.**

Below are the **key measures**:

1. Proper Planning

- Accurate forecast of **train movements, pathing, load availability, and power availability**

- Advanced planning of **empties**, **interchange**, and **terminal operations**
- Avoid unnecessary detentions at yards and stations.

✓ Better planning = smoother flow = higher throughput.

2. Effective Controlling (Real-Time Regulation)

- Intelligent regulation by **Section Controllers**
- Prioritizing trains to minimize **conflicts** and **detentions**
- Ensuring punctual departure from terminals
- Avoiding conflicting movements and unnecessary stops.

✓ Smart control = maximum utilisation of each train path.

3. Improving Operational Efficiency

- Speedier shunting
- Quick locomotive / crew change
- Efficient rake handling
- Minimizing avoidable terminal delays.

✓ Operational discipline directly increases sectional throughput.

4. Scientific Time-Tabling

- Preparing a timetable based on **practical line capacity**
- Balancing train paths (coaching vs goods)
- Removing conflicting timings
- Providing realistic sectional running times
- Scheduling maintenance blocks systematically.

✓ Well-planned timetabling gives **more usable paths**.

5. Proper Maintenance of Instruments

- Ensures reliability of:

- Block instruments
 - Point machines
 - Track circuits
 - Signals
 - Level crossing gates
 - Reduces failures → reduces train detentions.
- ✓ Fewer failures = higher usable capacity.
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6. Reducing Block Operation Time

- Quick release of block section
 - Efficient LC procedures
 - Prompt communication (SM ↔ Controller ↔ Loco Pilot)
 - Using standardised operational practices.
- ✓ Reduced “t” (block operation time) → **higher capacity** (as per Scott’s formula).
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7. Minimising Crossing and Precedence

- Avoid unnecessary crossing on single line
 - Regulate trains to prevent **slow train precedence**
 - Run “bunch-free” operation
 - Give paths based on **priority + running time compatibility**.
- ✓ Less crossing = less detention = more train paths.
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8. Implementing Management Information Systems (MIS)

- MIS helps in:
 - Monitoring running trains
 - Detecting bottlenecks
 - Analysing detention reasons
 - Optimising crew and locomotive utilisation.

✓ MIS = data-driven decisions → improved sectional performance.

9. Using New IT Software (Without Physical Cost)

- Real-time dashboards
- Crew management systems
- FOIS/TMS-integrated decision-making
- Yard management software.

✓ IT solutions optimise operations **without infrastructure cost**.

10. Adopting New Operational Techniques

- Skip-stopping operations
- Shunting-free trains
- Running sectional “convoys” where suitable
- Reducing manual processes

✓ Enhances sectional fluidity.

11. Avoiding Separate Traction for Different Sections

- Ensures:
 - No loco changeover delays
 - Common traction reduces operational slack
 - Continuous train movement without halts.

✓ Saves time → increases throughput.

12. Running Heavy-Haul Goods Trains

- Larger load per path → fewer trains required
- Reduces congestion
- Improves NTKM per path

✓ Same number of paths → higher carrying capacity.

METHODS OF IMPROVING LINE CAPACITY (BY INCURRING EXPENDITURE)

Line capacity can be significantly enhanced by **capital-intensive infrastructural and technological improvements**. These measures address speed, block length, signalling, yard capacity, and operational bottlenecks.

1. Reducing Running Time on Critical Block Section

- Improve permanent way
- Remove/reduce speed restrictions
- Upgrade track geometry
- Reduce block section length
→ Faster clearing of block = shorter headway = more trains.

2. Improving Speed of Trains

- Higher sectional speeds
- Increasing MPS (Maximum Permissible Speed)
- Upgrading tracks to higher standards
→ Higher speed = lesser occupation time of block = higher capacity.

3. Provision of Automatic Signals (ABS)

- Replaces manual/slot-based working
- Divides the section into **smaller automatic blocks**
- Reduces minimum headway
→ Capacity increases by **30–40%** on busy routes.

4. Use of Modern Signalling Systems

- Axle counters
- Electronic interlocking
- SSI / EI

- Panel Interlocking
→ Improves reliability, reduces failures, supports high-speed operations.

5. Provision of CTC (Centralized Traffic Control) on Double Line

- Remote control of points and signals
- Improved coordination
- Reduces human reaction time
- Faster regulation decisions
→ Boosts average speed and capacity.

6. IMPROVING THE SPEED OF GOODS TRAINS

7. Use of EMU/DMU for Stopping (Passenger) Trains

- High acceleration and deceleration
- Better turnaround
→ Reduces station dwell + reduces block occupancy time.

8. Provision of AWS (Automatic Warning System) / TPWS

- Enhances safety and visibility
- Allows closer train following in low-visibility
→ Works as a capacity-enhancer on fog-prone routes.

9. Provision of PI / RRI / EI

- Panel Interlocking (PI)
- Route Relay Interlocking (RRI)
- Electronic Interlocking (EI)
→ Faster signal setting, improved reliability, fewer failures
→ Reduces station clearance time, increasing section capacity.

10. Increasing CSL (Clear Standing Length) of Running Lines

- Lengthening loops, sidings, and platforms
 - Allows handling of longer trains (e.g., 58/61/72 BOXN)
→ Prevents train splitting → reduces detention → enhances capacity.
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11. Interlocking or Elimination of Level Crossing Gates

- Manned → interlocked → manned with boom → ROB/RUB
→ Reduces gate closing time
→ Fewer interruptions to train operations.
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12. Improving Facilities at Terminal Yards

- More reception & despatch lines
 - Extra pit lines
 - Better loco/crew facilities
 - Improved yard layout
→ Avoids terminal congestion → more trains can be operated.
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13. Provision of Longer Turnouts

- High-speed turnouts (1 in 18.5, 1 in 20, etc.)
 - Permit entrance/exit at higher speeds
→ Reduces station detention and increases throughput.
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14. Independent Shunting Facilities at Major Stations

- Dedicated shunting lines
 - Avoid interference with main line movements
→ Frees block sections → improves efficiency of reception/despatch.
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15. Easing of Ruling Gradient

- Reduces locomotive strain
- Increases hauling capacity

- Allows higher speed and fewer brake applications
→ Improves operational fluidity & capacity.

EXAM-ORIENTED SUMMARY (WRITE THIS IN ANSWER SHEET)

Line capacity can be enhanced by expenditure through measures such as:

- Reducing running time & improving speeds
 - Advanced signalling (ABS, EI/RRI/PI, CTC)
 - Modern systems (train describer, AWS/TPWS)
 - Longer loops, yard improvements, LC elimination
 - High-speed turnouts & independent shunting lines
 - Easing ruling gradient
- These collectively reduce headway, minimise operational delays, and increase safe train paths.

II. THROUGHPUT

1. Definition of Throughput

Throughput of a section is the **total quantum of traffic transported over a section in 24 hours**.

Traffic includes:

- **Passenger Traffic**, and
- **Freight Traffic**

2. Passenger Throughput (PKM)

Passenger throughput is calculated based on: NUMBER OF PASSENGERS TRAVELLED AND THE DISTANCE THEY TRAVELLED.

- Number of passengers travelled × Distance travelled

Formula

Passenger Throughput (PKM) = No. of passengers × Distance (km)

Measured in **Passenger-Kilometres (PKM)**.

5. Steps to Improve Throughput

Throughput can be improved in **two major ways**:

- (A) Running *more number of trains*
- (B) Increasing *load per train*

A. Running More Number of Trains

Key measures

- a) Utilize section capacity fully; run trains with **CC (Commodity Classification)** loads
- b) Avoid Light Engine movements—**ensure every loco hauls adequate load**
- c) Adopt strong **marketing strategies** to generate more traffic
- d) **Timely ordering of trains** by Control
- e) Set periodic **loading targets** and monitor performance
- f) Incentivize efficient staff (Crack Specials, premium freight specials)
- g) Switch from **Diesel → Electric traction** to improve sectional speed
- h) Deploy **higher horsepower locos** to run faster, heavier trains

B. Increasing Load per Train

Measures

- a) Increase **loop capacities**
 - b) Introduce **high-capacity wagons**
 - c) Improve and strengthen **track structure**
 - d) Use **Multiple Locomotive Operation (MLO)** or more powerful locomotives
 - e) Use **high-capacity bearings** on rolling stock
 - f) Reduce **tare weight** of wagons
 - g) Use **locator technology / tracking systems**
 - h) Use **low-platform bogie containers** to maximize payload
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